Data Retrieval with get method

In Hibernate, the get() method is used to retrieve an entity from the database based on its primary key. It fetches the data and converts it into a Java object automatically. Here's a detailed explanation of how data retrieval works using Hibernate's get() method:

1. Entity Definition (Student Class)

First, we define the entity class that is mapped to the database table. In this case, the class Student is mapped to the StudentTable in the database.

```
import jakarta.persistence.Column;
import jakarta.persistence.Entity;
import jakarta.persistence.Id;
import jakarta.persistence.Table;

@Entity
@Table(name="StudentTable")
public class Student {
	@Id
	@Column(name="sid")
		private int id;

	@Column(name="sname")
		private String name;

	@Column(name="scity")
		private String city;

		public Student() {
				System.out.println("Zero param constructor");
		}
```

```
    public int getId() {
        return id;
    }
    public void setId(int id) {
        this.id = id;
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getCity() {
        return city;
    }
    public void setCity(String city) {
        this.city = city;
    }
}
```

2. Data Flow During Retrieval

Here's how data retrieval happens step-by-step using get():

- **Step 1**: The get() method is called on the Session object. For example:

```
Student student = session.get(Student.class, 1);
```

- **Step 2**: Hibernate (ORM) takes care of the following:
    - It reads the **mapping details** from annotations (@Entity, @Table, @Column) to understand which table and columns to target.
    - It uses the configuration settings for the database connection.
- **Step 3**: Hibernate translates the get() method call into a SQL query using **JDBC**:

```
SELECT * FROM StudentTable WHERE sid=1;
```

- **Step 4**: **JDBC** executes the query on the database, retrieving the record.
- **Step 5**: Hibernate processes the result set. It finds the matching fields from the result set (sid, sname, scity) and maps them to the corresponding fields in the Student class.
- **Step 6**: Hibernate automatically creates an instance of the Student class and populates the fields with the data from the result set.
- **Step 7**: Hibernate calls the **zero-parameter constructor** (Student()) of the entity class and sets the values using setter methods (setId(), setName(), setCity()).

3. ResultSet to Object Mapping

Internally, Hibernate uses **ResultSet** from JDBC to get the data from the database. For each record in the ResultSet:

- It **calls the constructor** of the Student class to create an object.
- It **sets the values** of each field (id, name, city) using reflection and setter methods.

4. Main Class to Demonstrate Retrieval

Here's a simple main class that demonstrates how to retrieve data using Hibernate's get() method.

```
import org.hibernate.Session;
import org.hibernate.SessionFactory;
import org.hibernate.cfg.Configuration;

public class Main {
        public static void main(String[] args) {
        // Configure Hibernate
        SessionFactory factory = new
Configuration().configure("hibernate.cfg.xml")
                .addAnnotatedClass(Student.class).buildSessionFactory();

        // Open session
```

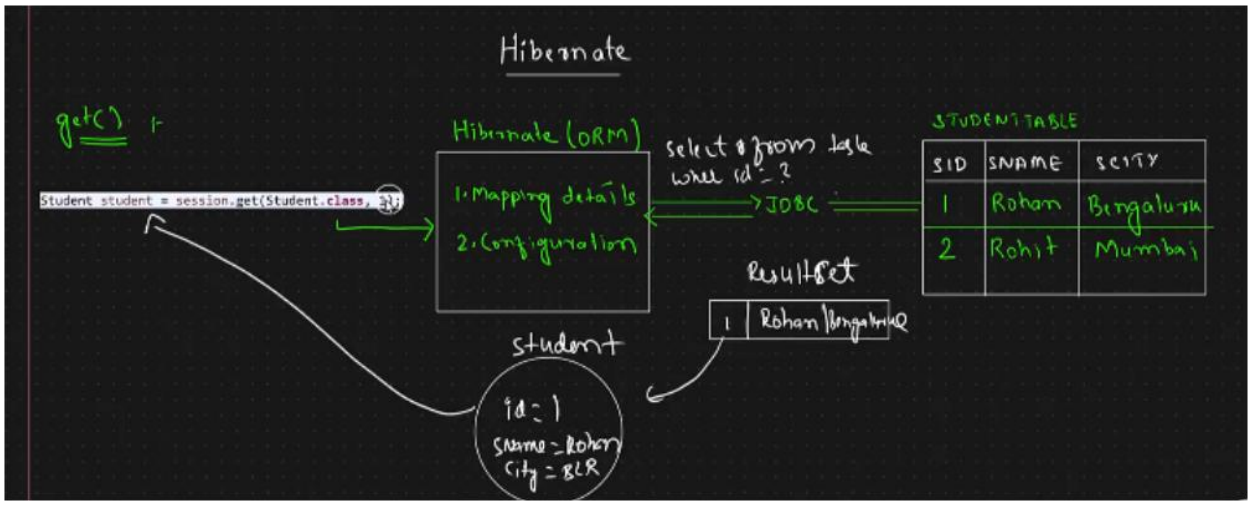
```
        Session session = factory.openSession();

        try {


        // Retrieve student with ID 1
        Student student = session.get(Student.class, 1);

        // Display the student details
          System.out.println("ID: " + student.getId());
          System.out.println("Name: " + student.getName());
          System.out.println("City: " + student.getCity());


        } finally {
        // Close session
          session.close();
          factory.close();
        }
        }
}
```



5. How ORM Works Behind the Scenes

- Hibernate simplifies the database interaction by mapping Java objects (entities) to database tables.
- The get() method abstracts away the complexity of SQL queries and JDBC code.
- **Automatic Object Creation**: Hibernate takes care of object creation from database records. You don't have to manually write code to parse the ResultSet.
- **Zero-Parameter Constructor**: Hibernate requires a no-arg constructor to create instances of the entity class.

6. Advantages of get() Method

- **Simple API**: The get() method is simple to use and returns null if no entity is found.
- **Eager Loading**: The get() method fetches the object immediately and returns a fully initialized entity.
- **Convenience**: Hibernate handles mapping and data retrieval without manual SQL queries.